

The Evolution of Identity Signals for Coordination in Diverse Societies

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Abstract

Individual social identities indicate group affiliations and are typically associated with group-typical preferences, signals that indicate group membership, and the propensity to condition actions on the social signals of others, resulting in group-differentiated interaction norms. Past work modeling identity signaling and coordination has typically assumed that individuals belong to one of a discrete set of groups. Yet individuals can simultaneously belong to multiple groups, which may be nested within larger groupings. Here, we introduce the *generalized Bach or Stravinsky game*, a coordination game with ordered preferences, which allows us to construct a model that captures the overlapping and hierarchical nature of social identity. Our model unifies several prior results into a single framework, including results related to coordination, minority disadvantage, and cross-cultural competence. Our model also allows agents to express complex social identities through multidimensional signaling, which we use to explore a variety of complex group structures. Our consideration of intersectional identities exposes flaws in naive measures of group structure, illustrating how empirical studies may overlook some social identities if they do not consider the behaviors that those identities function to afford.

Keywords: identity signals; intersectionality; coordination; Bach or Stravinsky; cultural evolution

Social Media Summary

We develop a formal model of the evolution of multidimensional social identity signals, allowing us to explore the emergence of complex group structures, cross-cultural competence, and intersectional identity in diverse societies.

1 Introduction

As early as the Third Dynasty of Ur (2100 BCE) in ancient Mesopotamia, we have evidence of complex social identities; cities as far apart as Ur and Mari, well over 100 miles apart, evidence a body of literature indicative of a shared Sumerian identity, while literature concerned with each city’s patron gods indicate more localized social identities (Delnero, 2016). The layered and overlapping identities of this case are indicative of similar stories in human history, as ethnographic and archaeological evidence suggests that coalitions, warfare, fusion-fission dynamics, and migration are features of early human groups (Singh and Glowacki, 2022). As a result, humans likely used a large variety of markers of group identity to facilitate coordination within and between groups (Hodder, 1982; Cohen, 2012; Johnson et al., 2024; Smaldino, 2019), especially in groups too large for a group member to know every individual in the group (Moffett, 2013). In today’s large, diverse, and globally-connected societies, complex group identities are commonplace (Roccas and Brewer, 2002).

Some social groupings follow straightforward principles of organization. For example, all astrophysicists are physicists and all physicists are scientists. As such, you can assume that all astrophysicists share things in common with all physicists and all physicists share things in common with all scientists. But there are also ways in which social identities break from the contours of simple set membership relations. For example, the mere conjunction of dominant narratives in Black liberation and in mainstream feminism does not yield important narratives of Black feminists (Combahee River Collective, 1977; Crenshaw, 1991).

To get at this richer understanding of identity and power dynamics, we will consider how people come to optimize interactions in populations with diverse and multifaceted sets of preferences. Some prior literature, considering the power dynamics of how people are related to each other, has referred to such sets of identity relationships as group topologies (Burt, 1977; Bourdieu, 1985; Ioannides, 2006). We use the more generic term “group structures” to refer to the relationships between preferences, signals, and interpersonal behaviors that emerge in populations of interacting individuals. This paper provides a formal model of how a variety of social identity structures can obtain in a population of agents trying to coordinate behavior and preferentially interact with ingroup members.

1.1 Preferences, signals, and actions

Conceptually, there are three aspects of social group identities that we model: (i) the *preferences* of individuals, which correlate with social group membership, (ii) the emergence of *social signals* used to express and identify group membership, and (iii) the use of those social signals to *coordinate actions*. In high school cliques, Goths and Emos may have different values and music preferences; they use differences in hair and clothing style to identify group members, which helps them decide who to join at recess. Swingers might use upside-down pineapples to signal their preferences for “coordinating behavior” with other couples (Pelzer, 2023). Similar empirical examples abound (Barth, 1969; Berger and Heath, 2008; Lin et al., 2024). But, these examples still lack the complexity that we often exhibit in our social identities, in which preferences for how we behave and interact are weighed against the availability of suitable partners, leading those aspects of expressed identity to be highly context-dependent (Roccas and Brewer, 2002; Smaldino, 2019; Bunce, 2020; McCluney et al., 2021). Our model is intended to create continuity between prior work on identity signaling and coordination that has dealt with simple discrete identities (Boyd and Richerson, 1987; Nettle and Dunbar, 1997; McElreath et al., 2003; Castro and Toro, 2007; Smaldino et al., 2018; Smaldino and Turner, 2022; Smaldino, 2022; Goodman et al., 2023; Macanovic et al., 2024) and the need to capture these more complex, context-dependent identity-mediated interactions.

Our model assigns agents different preferences reflective of a complex society and illustrates how these preferences can lead to social signaling of complex social identities as well as norms for coordination that are conditioned on the signals that agents perceive. For convenience, we hold agent preferences fixed, and allow strategies for social signals and coordinative actions to evolve. In reality, preferences may also evolve; this aspect of our model coheres with much of the prior literature and is a convenient simplifying assumption. In Appendix B, we discuss an alternative learning dynamic for the model that can facilitate future research adding dynamically evolving preferences to the model. We use the term ‘group structure’ to refer to the union of agents’ preferences, signaling strategies, and behavioral strategies for interacting with others contingent upon the receipt of particular signals. Group structures are an emergent feature of our model. When referring to a subset of the population that shares the same preferences, we use the terms ‘preference type’ or ‘preference group’ interchangeably.

1.2 The Mesopotamian Approach

Our approach diverges from most prior models in two important ways. First, we allow agents to broadcast social signals in multiple dimensions rather than a single dimension. Second,

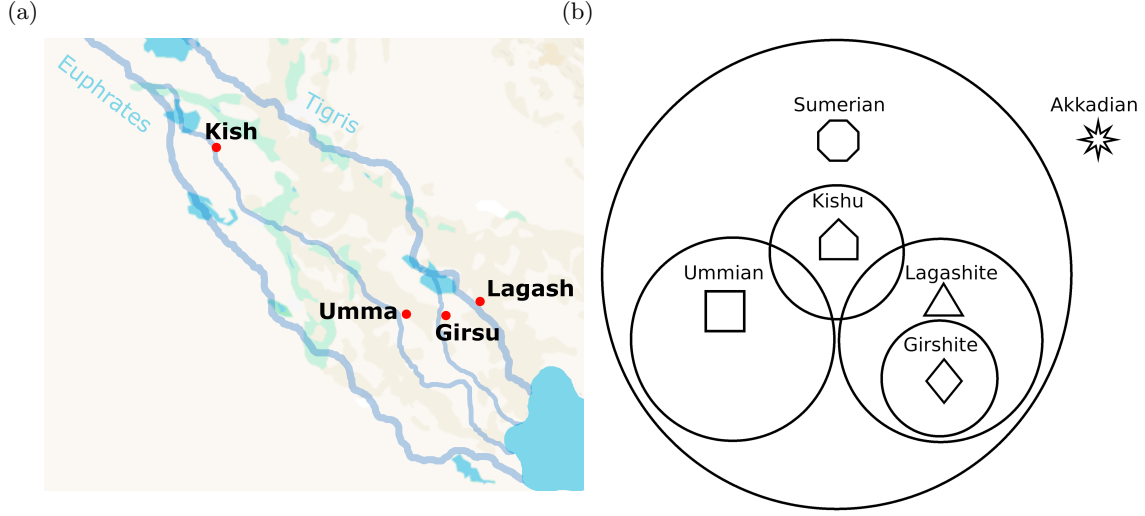


Figure 1: (a) Ancient Sumerian cities used throughout the paper as a motivating example. (b) Diagram of relationships between the different Mesopotamian identities along with shapes used to represent each of them throughout the paper.

we allow a broad range of payoffs that can be associated with successful coordination, based on the preferences of the individuals involved. Consider a hypothetical example in which Catholics prefer to greet others with handshakes while Protestants prefer hugs. If a Catholic and a Protestant end up hugging upon meeting, the Protestant receives a higher payoff than the Catholic, but both receive higher payoffs than if they had entirely failed to coordinate on a greeting. We will continue to use the example of greetings as a way of describing coordinating behavior, though our model can represent any coordinative activity.

To avoid having to construct artificial identity structures to motivate our analysis, we will anchor our discussion of the model with an example of social groups from ancient Mesopotamia (Figure 1). This is a narrative device; the specific preferences and actions we assign to each group of agents are not necessarily reflective of historical reality. In ancient Mesopotamia there were two rival Sumerian cities, Umma and Lagash, who disagreed continually over who would control the fertile land between them. Let us assume that members of each group preferred the greeting of their own city over the greeting of the other. Despite their rivalry, both were Sumerian speaking cities, and so we assume that members of both groups preferred a Sumerian greeting over an Akkadian greeting. The Akkadians were a Semitic language-speaking people living in and around northern Sumeria, who eventually conquered the region in 2334 BCE. Prior to that, Lagash established the city of Girsu in the middle of the disputed territory between themselves and Umma, where the temple of Lagash’s patron deity Ningirsu was housed. We can therefore think of Girshites as being a rather zealous subgroup of the broader Lagash community. Finally, let us consider the Sumerian city of Kish. Around 2550 BCE, the king of Kish negotiated a treaty between Umma and Lagash. Thus, we can treat Kishus as having some overlapping identity with Ummians, and perhaps with the Lagashites as well.

In our analyses, a large population of agents plays a correlative coordination game that allows agents to coordinate despite having different preferences. Section 2 introduces the baseline model, including the generalized Bach or Stravinsky game, the learning dynamics

that allow strategies to evolve, and the results of simulations considering how coordinating behaviors evolve in the absence of signals. Section 3 explains how (potentially costly) social identity signals are incorporated into the model, and presents simulation results for simulations in which agents can learn to use unidimensional signals to condition their behavior upon. These simulations establish a few properties of the model that are carried forward as more complex (multidimensional) signaling is added. First, we find that larger communities who share the same preferences are privileged over smaller preference groups; specifically, the population as a whole is more likely to settle on the preferences of a larger preference group than a smaller preference group. Second, we find that the presence of a larger preference group incentivizes other smaller preference groups to adopt social identity signals that they might not otherwise use. Finally, in Section 4, we consider the emergence of complex group structures involving coordination among nested or overlapping preference types (Figure 2), made possible by allowing agents to signal in multiple dimensions simultaneously.

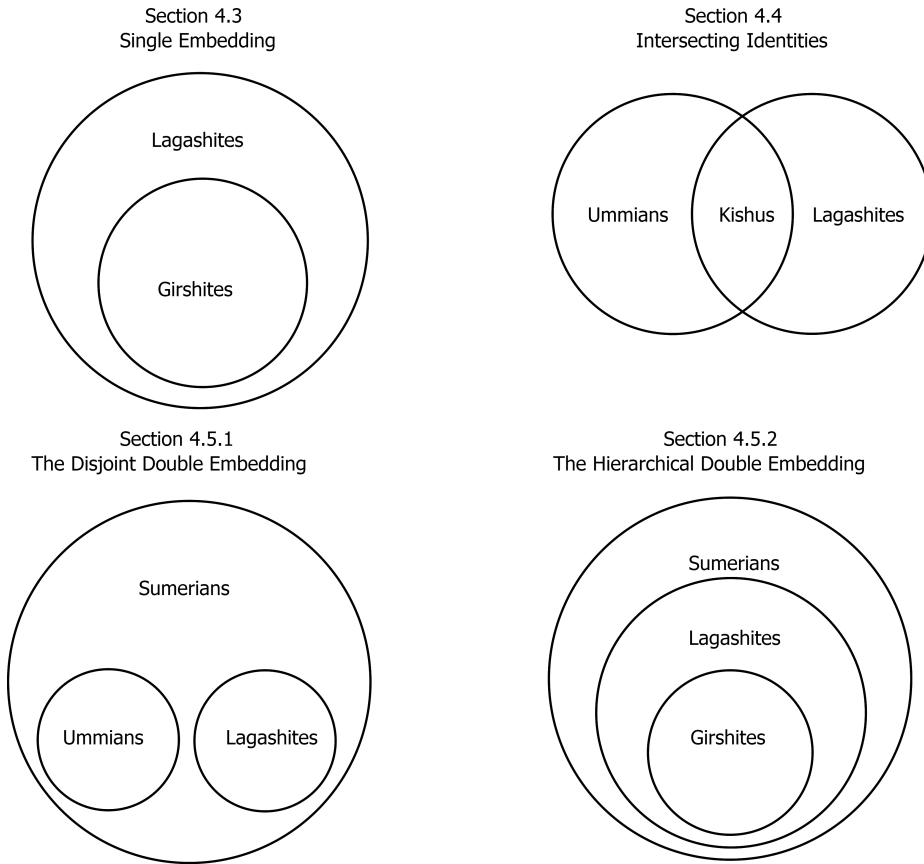


Figure 2: A rough sketch of some group structures that can be produced with our model.

2 Base Model Description and Signal-Free Results

Our model is complex, but it has simple roots. In this section, we introduce the base model, which includes agent interaction and learning, but does not yet include identity signaling or signal-contingent behavior. We then present simulation results indicating the model outcomes in the absence of signaling. Python code for all model versions presented in this

paper is available here: https://github.com/nathanlgabriel/social_identity_signaling

2.1 Generalizing the Bach or Stravinsky Game

The traditional Bach or Stravinsky (BoS) game is a one-shot, two-option coordination game between two players. It was introduced to the literature by Luce and Raiffa (1957) as the “battle of the sexes” game, but has since been re-branded the “Bach or Stravinsky” game to dissociate it from undesirable prejudices and stereotypes while maintaining the same abbreviation, BoS (Dekker and Van Rooy, 2000). In this game, each player prefers to coordinate on a different behavior—one would rather go to the Bach concert, the other would rather hear Stravinsky—but both players prefer *some* coordination to non-coordination. The payoff matrix is as follows:

		Player 2	
		Bach	Stravinsky
Player 1	Bach	$1 + \alpha, 1$	$0, 0$
	Stravinsky	$0, 0$	$1, 1 + \alpha$

Table 1: Traditional Bach or Stravinsky Game; $\alpha \geq 0$. Each cell lists the payoffs to players 1 and 2, respectively.

The game captures a broad class of scenarios in which individuals have different preferences for coordination. We coordinate when deciding to shake hands or high five, or when choosing music for a party. Such coordination problems are commonplace in communication; e.g. the same thing might be called “battle of the sexes” or “Bach or Stravinsky”, it might be called “the morning star”, “the evening star”, or “Venus”; it might be called “the electric slide” or “the wobble”.

To generalize the BoS game, we first consider the scenario in which the game is played repeatedly among intermixing agents in a population, such that all three dyadic preference combinations are possible. Agents’ payoffs in the game are given by their preference type: Bachites prefer Bach, Stravinskians prefer Stravinsky. Thus, the game is characterized by three payoff matrices, representing each possible pair-up:

		Bachite	
		Bach	Stravinsky
Bachite	Bach	$1 + \alpha, 1 + \alpha$	$0, 0$
	Stravinsky	$0, 0$	$1, 1$
		Stravinskian	
		Bach	Stravinsky
Bachite	Bach	$1 + \alpha, 1$	$0, 0$
	Stravinsky	$0, 0$	$1, 1 + \alpha$
		Stravinskian	
		Bach	Stravinsky
Stravinskian	Bach	$1, 1$	$0, 0$
	Stravinsky	$0, 0$	$1 + \alpha, 1 + \alpha$

Table 2: Generalized Bach or Stravinsky for a population of two types; $\alpha \geq 0$. Each cell lists the payoffs to each player: first the row player, then the column player.

This generalization of the Bach or Stravinsky game is equivalent to Neary (2012)’s Language Game, at least when restricted to two types of players with two actions and no social signals.

Since agents always get a payoff of 0 when they fail to coordinate, we can more succinctly present agents’ payoffs in terms of their preferences for successfully coordinating on each possible behavior:

Coordination Preferences	Bach	Stravinsky
Bachite	$1 + \alpha$	1
Stravinskian	1	$1 + \alpha$

Table 3: Coordination Preferences: generalized BoS for a population of two types, $\alpha \geq 0$.

Table 3 expresses all of the information in Table 2, but does so much more concisely. This concise format will be particularly useful when we consider populations with more than just two preference types and more than two actions.

2.2 Evolutionary Dynamics

In our first simulations, we initialized a population of $N = 1,000$ agents with a fixed distribution of preference types and randomly assigned a behavior strategy to each (for the two-option BoS game, each player was initially assigned a strategy of playing either Bach or Stravinsky without regard for preference). In all extensions of the model, we always begin simulations with agents whose strategy profiles are randomly assigned (Appendix H explores a version of the model with non random initial conditions and finds similar results to the simulations with random initial conditions). The model dynamics proceeded in discrete timesteps, using replicator dynamics under the assumption of a large number of interactions in a well-mixed population, such that agents who acquired higher payoffs were more likely to transmit their strategies. These generic replicator dynamics allow us to remain uncommitted to the nature of the transmission mechanism or the timescale thereof—it could represent vertical parent-to-offspring transmission through genes or teaching, or success-biased social learning of older or same-age individuals (Smaldino, 2023). As detailed below, we think it is most intuitive to view this as payoff-biased oblique social transmission.

On each timestep, the prevalence of each strategy profile among a given preference type k is adjusted according to a discrete replicator equation:

$$N_{k,t+1}(x) = N_{k,t}(x) + N_{k,t}(x) \times [U_k(x) - \text{Avg}(U_k(i))_{i \in X}]$$

where, for a given preference type, X is the set of all strategy profiles, $N_{k,t}(x)$ is the number of agents of type k with strategy profile x at time t , $U_k(x)$ is the utility (expected payoff) of strategy profile x , and $\text{Avg}(U_k(i))_{i \in X}$ is the average utility of all the strategy profiles present among the given type.

The utility of strategy profile x for preference type k is calculated as:

$$U_k(x) = \begin{cases} \sum_{i \in Y} [M(i) \times p_{xi,k}] & \text{if } x \neq i \\ \sum_{i \in Y} [(M(i) - 1) \times p_{xi,k}] & \text{if } x = i \end{cases}$$

Where Y is the set of all strategy profiles present in the population, $M(i)$ is the number of agents (of any preference type) in the population who play strategy i , and $p_{xi,k}$ is the payoff to an agent of preference type k for playing strategy x when paired with an agent

who plays strategy i . After adjusting the prevalence of strategy profiles, their quantity is re-normalized so that the number of agents of each preference type remains constant throughout a simulation. Thus, utility is calculated based on the assumption that an agent interacts with a large representative sample of all the other agents in the population.

The replicator dynamics perhaps most intuitively approximate agents inheriting their strategy profiles from among the highest performing agents of the previous generation. This does not require that agents can observe everyone’s preference type or strategy profile. Rather, it is justified by basic life history assumptions. Individuals are likely to learn their behavioral strategies during childhood, during which time most of the observed elders will be part of their local family groups and affiliative communities, and therefore likely to share their preference type. Evolutionary modeling suggests that such intra-group inheritance is often adaptive in diverse populations (Bisin et al., 2011; Smaldino and Velilla, 2025), and as implemented here only requires that payoffs (or indicators thereof, such as prestige) are observable. Meanwhile, payoffs are accrued during adulthood, during which time individuals are assumed to interact more widely with members of the entire population. The effects of more structured interaction networks are postponed for future research. In Appendix B, we present analyses in which individuals agents learn their strategy profiles via reinforcement learning rather than having those strategies evolve via replicator dynamics. We obtain qualitatively similar results, further adding to the robustness of our analyses.

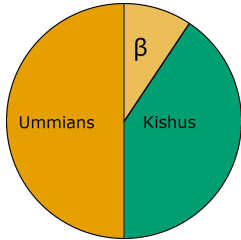
We also allow for the possibility that an agent’s strategy profile is not copied faithfully, or “mutates.” This is governed by two parameters, m_0 and m_1 . Each agent is selected for mutation with probability m_0 . If selected, each element in the string expression of the agent’s strategy profile is assigned a random value from among the allowable values with probability m_1 . In the model described thus far, agents’ strategy profiles are represented by strings of length one, and can either be $\langle B \rangle$ (play Bach) or $\langle S \rangle$ (play Stravinsky). As the model is extended to allow agents to broadcast social signals and to choose their action based on the signal of a paired agent, the string expression of agents’ strategy profiles will have length greater than one. Basic model parameters used in our simulation results are given in Table 4.

	Number of simulations	Simulation length, T	Population size, N	m_0	m_1	Signal cost, c
Section 2.3 Figure 3	10,000	4×10^4	1,000	0.01	0.1	n/a
Section 3.2 Figure 4	10,000	4×10^4	1,000	0.01	0.1	n/a
Section 3.4 Figures 5 & 6	1,000	8×10^4	1,000	0.01	0.1	0.01
Section 4.2 Figure 9	1,000	4×10^4	1,000	0.01	0.1	0.0005
Section 4.3 Figure 11	1,000	4×10^4	1,000	0.01	0.1	0.0002
Section 4.4 Figure 12	100	4×10^4	1,000	0.05	0.2	0.0002
Section 4.4 Figure 13	100	4×10^3	1,000	0.05	0.2	0.0005

Table 4: Simulation parameter values used for the results presented throughout this paper.

2.3 Simulation Results: Majority Groups Have Power to Make Their Preference the Norm

We present our results using the Mesopotamian framing introduced earlier. We consider a population with two types of agents, Ummians ($\square$) and Kishus ($\triangle$). Ummians ($\square$) make up a proportion $0.5 + \beta$, with Kishus ($\triangle$) constituting the remainder. Coordination preferences follow the generalized BoS game, such that members of each preference group receive a payoff of $1 + \alpha$ for coordinating on their preferred greeting, 1 for coordinating on their non-preferred greeting, and zero for failing to coordinate (Table 5).



Coordination Preferences	Umma greeting	Kish greeting
Ummians $\square$	$1 + \alpha$	1
Kishus $\triangle$	1	$1 + \alpha$

Table 5: Payoffs for the baseline model with Ummians and Kishus.

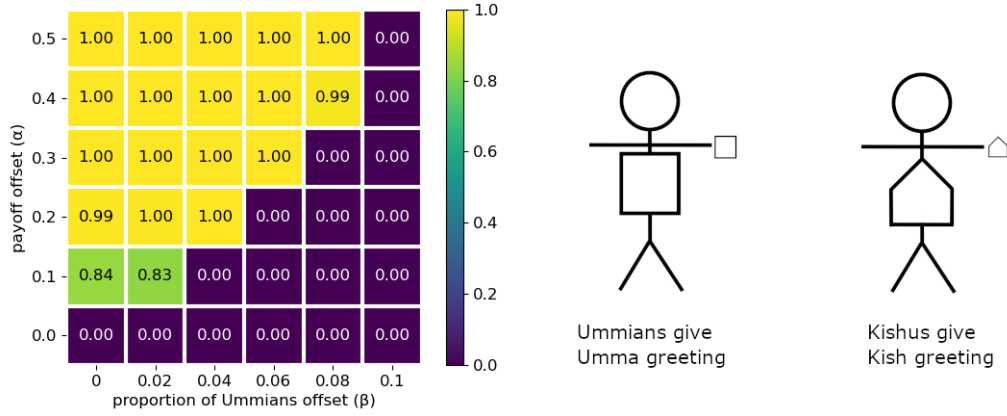
Our simulations results are illustrated in Figure 3. In the absence of social signaling, agents will evolve to simply use whatever strategy yields the highest expected payoff. Consequently, there are only two prominent types of outcomes in the simulations. When preferences are weak (α is small), everyone converges to using the same greeting because the benefit of always coordinating outweighs the benefit of always giving one's preferred greeting but frequently failing to coordinate behavior. It is important to note that our model is a stochastic system that can exhibit some path dependency. So a particular set of parameters, say $\alpha = 0.1$ and $\beta = 0$, might result in 84% of simulations reaching the outcome depicted in Figure 3(a), while other simulations with those same parameters resulted

in different outcomes.

If the two preference groups are of unequal size, the equilibrium greeting will typically be that the majority preference group (the Ummians in this case). Correspondingly, when β is large, the population will converge to all use the Ummian greeting for larger values of α , because for the Kishus, the higher payoff for coordinating on their preferred greeting is outweighed by the higher probability of interacting with an Ummian (see also Neary, 2012; O'Connor, 2017). On the other hand, when preference groups are more evenly sized ($\beta \approx 0$) or coordinating on one's preferred greeting is strongly incentivized (α is large), each preference group will evolve to use their preferred greeting. When $\alpha = 0$, the population evolves to coordinate on either greeting about half the time, as either coordinative equilibrium is equally good. In Appendix E, we show how these results can be derived analytically.

It may be helpful to think of this model as having three basins of attraction, one for each of the three equilibria. These equilibria are: one in which each type plays their preferred greeting (Figure 3 (a)), one in which everyone gives the Ummian greeting (Figure 3 (b)), and one in which everyone gives the Kishu greeting (not depicted, but can be inferred as occurring whenever the other two equilibria do not obtain). Because each simulation was initialized with randomly assigned strategy profiles, the proportion of simulations runs leading to each equilibrium can be interpreted as the relative size of each basin of attraction.

(a) Proportion of simulations in which types evolve to use their preferred greeting.



(b) Proportion of simulations in which everyone evolves to use the Ummian greeting.

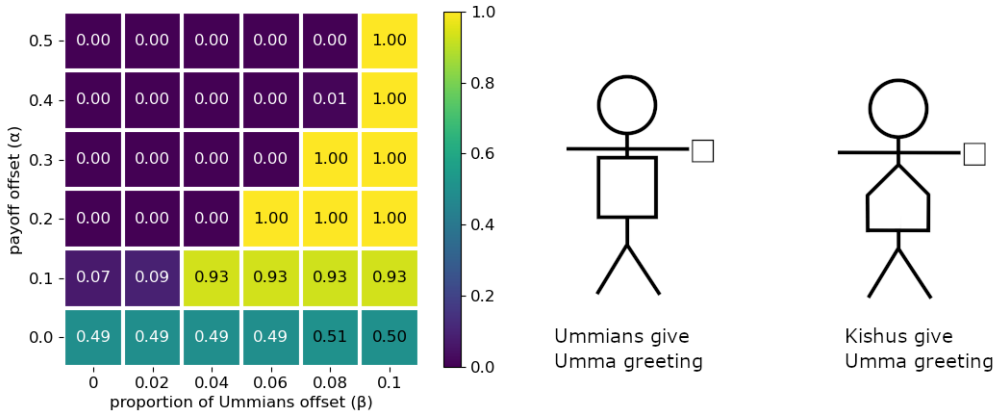


Figure 3: (a) Proportion of simulations in which agents always give their preferred greeting. (b) Proportion of simulations in which everyone gives Ummians' preferred greeting.

It is important to notice just how undesirable this behavior is for agents. When agents of each type evolve to use their preferred greeting, they fail to coordinate (and receive a payoff of zero) every time they interact with an agent of another type. And when agents all evolve to use the greeting of the majority preference group (the Ummians), interactions between two members of the minority preference group (the Kishus) always yield a lower payoff than would occur if both individuals were to use their preferred greeting. This is why, in part, cultures have evolved to use social signals and ethnic markers (McElreath et al., 2003; Smaldino, 2019, 2025). As we will show in the following section, social signals allow people to condition their actions on those signals such that they successfully coordinate actions when interacting with individuals with differing preferences and coordinate on their preferred action when interacting with individuals who share their preferences. Even when a majority preference group makes their preference the norm, minority preference groups can still establish their own practices to use within their preference group.

Once a preference group becomes correlated with a distinct social signal (or combination of social signals) as well as a distinct way of acting in response to social signals, it makes sense to refer to the members of the preference group as a social group in the ordinary sense. Indeed, we think it would make sense to call the Ummians and Kishus separate groups in this Section's model when they are identifiable by their actions, i.e. when the

outcome depicted in Figure 3a arises. However, when Ummians and Kishus look and act the same, as in the Figure 3b outcome, it seems odd to refer to them as separate groups. In this case, Ummians and Kishus are only differentiable from the omniscient modeler’s perspective. From the agent’s perspective, everyone looks and acts the same, and no one can tell that not everyone shares the same preferences.

3 Signals, Signaling Costs, and Attention

3.1 Adding Signals to the Generalized BoS Game

We next allow agents to broadcast social signals at the start of an interaction and to condition their action in the game upon their partner’s signal. Agents’ strategy profiles can be expressed as strings of length $1 + \theta$, where θ is the number of potential signals that can be received. The first element in this string represents the social signal that an agent broadcasts and the subsequent elements represent the action that is played when paired with an agent who broadcasts the corresponding signal. For example, suppose that there are two signals $\{1, 2\}$ in the two-option BoS game. Then there are 2^3 strategy profiles: $\langle 1BB \rangle$, $\langle 1BS \rangle$, $\langle 1SB \rangle$, $\langle 1SS \rangle$, $\langle 2BB \rangle$, $\langle 2BS \rangle$, $\langle 2SB \rangle$, and $\langle 2SS \rangle$. The strategy profile $\langle 1BS \rangle$ corresponds to an agent broadcasting social signal 1, playing Bach when paired with an agent who broadcasts 1, and playing Stravinsky when paired with an agent who broadcasts 2. Note that individuals in this model broadcast their signals overtly without first knowing anything about their audience; we do not take into account covert or strategic signaling (Smaldino et al., 2018; Smaldino and Turner, 2022). Successful individuals in this extension of the model transmit their entire strategy profile, with mutation as described in the previous section.

3.2 Simulation Results: Social Signals Allow Optimal Behavior and Cross-Cultural Competence

We continue with the example of Ummians ($\square$) and Kishus ($\triangle$) introduced in the previous section (Table 5), with the difference that agents now broadcast one of two social signals (orange or blue) and choose their action based on the signal they observe from a partner. We continue with the assumption that agents are paired at random and use signals to condition their behaviors. Agents do not use signals assortatively to find interaction partners as in some previous models (e.g., McElreath et al., 2003; Castro and Toro, 2007; Smaldino et al., 2018). We view this assumption as reasonable, as individuals must often condition their behaviors on information received after an interaction has already begun. We analyze a version of our model with this type of assortment in Appendix A, and show that the results are qualitatively similar.

Roughly speaking, an optimal outcome occurs when the highest net payoffs possible, given the distribution of players’ incentives, have evolved. In these initial extensions of the model, we see that social signals allow agents to successfully coordinate in contexts where they frequently failed to coordinate in the absence of social signals (e.g. Figure 3a). Social signals also allow agents to coordinate on their most preferred actions instead of on a less-preferred action (e.g. Figure 3b). After we have fully described all components of the model, Section 4.2 gives explicit criteria for what we designate as an optimal outcome; this later definition is consistent with what we call optimal in this section. For now, we emphasize that our use of ‘optimal’ is not reflective of anything like an ethically prescribed

group structure. For example, we sometimes find that the preferences of majority groups become normative for all inter-group interactions (e.g., Figure 4b, Figure 6). We do not think this is optimal in any prescriptive sense.

Broadly described, our analyses show that individuals almost always learn to use signals for optimal coordination, illustrating an often theorized use for identity signals (Boyd and Richerson, 1987; McElreath et al., 2003; Smaldino, 2019, 2025). When Ummians interact with other Ummians, they signal their identity and give the Umma greeting. Likewise, when Kishus interact with other Kishus, they signal their identity give the Kish greeting. But when Ummians interact with Kishus, the population still has to settle on whether to coordinate on the Umma greeting or the Kish greeting. The majority population is still advantaged in this case. If the Ummians outnumber the Kishus, then the population is more likely to settle on the Umma greeting as the norm when people of different types interact.

There are two types of optimal outcomes in this model, each of which involves agents successfully coordinating on their preferred greeting when interacting with partners of the same type, and otherwise settling on one of the two available greetings when interacting with partners of another type. Figure 4a shows the proportion of simulations that resulted in optimal outcomes of either type, and Figure 4b shows the proportion of simulations resulting in the subset of these outcomes favoring Ummians ($\square$). In Figure 4a we see that simulations more reliably converged to optimal outcomes when agents were increasingly incentivized to coordinate on their preferred greetings (larger α). In Figure 4b we see that increasing β (the proportion of Ummians) made outcomes favoring Ummians more frequent, just as in the no-signal case. When both types were present in equal proportions ($\beta = 0$), each optimal outcome was equally likely to occur. In the remaining proportion of simulations (1- the value in Figure 4a), meaningful signals did not evolve and either all agents in the population always gave the Ummian greeting irrespective of the social signal seen or all agents in the population always gave the Kish greeting irrespective of the social signal seen. No simulation resulted in each type of agent always giving their preferred greeting and then failing to coordinate when interacting with agents of a different type, as was frequently the case in the base model.

These results support similar findings from earlier models. Prior modeling work (O'Connor, 2019; Bunce and McElreath, 2018) has suggested that the ability of minority groups to maintain their cultural norms is tenuous at best, which accords with our results from the no-signal model in the previous section. Using a more complex model than Bunce and McElreath (2018), however, Bunce (2021) demonstrated that a minority group can nevertheless maintain their cultural norms if they are able to competently adopt either the majority or minority norms depending on the context, which Bunce calls cross-cultural competence. Our model also demonstrates the emergence of this sort of cross-cultural competence when agents are able to condition their actions on identity signals (Figure 4b). Our model complements Bunce (2021)'s account of cross-cultural competence by allowing an extra degree of freedom where social signals and conditional actions can evolve independently rather than assuming the existence of a single phenotype that packages each of these components together.

While our simulations frequently resulted in optimal outcomes, as shown in Figure 4a, they did not *always* result in optimal outcomes. When a simulation did not result in an optimal outcome, it was always the case that every agent in the population was giving the same greeting all of the time. This helps illustrate two important points. First, our simulations are stochastic. Our model shows that under certain conditions—e.g. when there are two different preferred ways to coordinate and there is an available means of

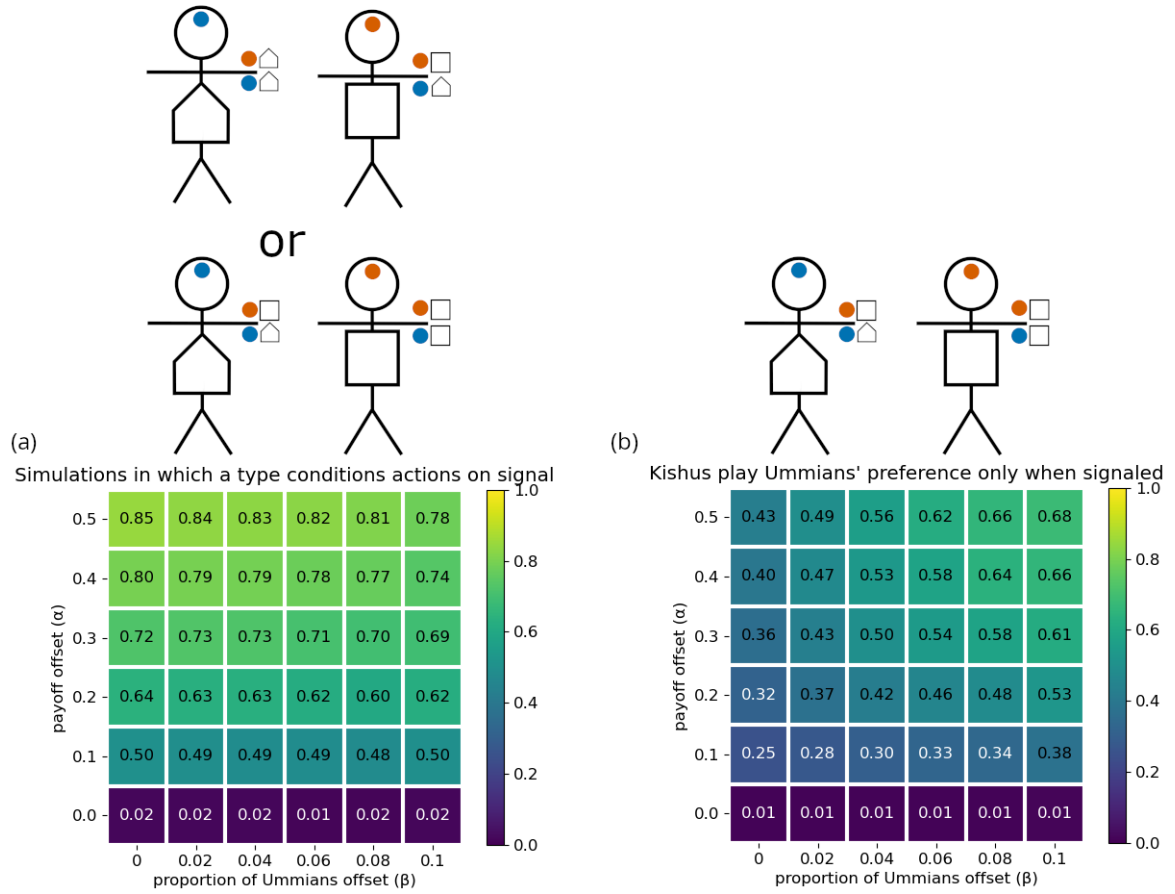


Figure 4: (a) Proportion of outcomes that are one of the two optimal outcomes. (b) Proportion of outcomes that are the optimal outcome favoring the Ummians.

signaling identity—agents are *likely* but not guaranteed to adopt social signals to facilitate coordinating in their preferred way when they are among others who share their preferences. Second, it shows clearly how group structure is not hard-coded but rather emerges in our model.

3.3 Adding Attention and Signal Costs to the Generalized BoS Game

Thus far, we have assumed that the production of signals and the ability to attend to and condition actions upon those signals is trivial and costless. However, it is often the case that social signals require some effort to broadcast or attend to. In such cases, only those groups for whom the social information is relevant should be expected to invest the effort to engage with the signals. For example, individuals holding fringe or extreme political opinions may invest much more effort to differentiate themselves from the mainstream, while those in the mainstream pay little heed to those same distinguishing characteristics (van der Does et al., 2022). To capture this, the model is extended to include a special signal, 0, which indicates an agent is not attending to signals, and a signal cost, c , that an agent incurs if she broadcasts any signal other than 0. When an agent does not attend to signals (i.e., when she broadcasts 0), she interacts with all other agents as if they had also broadcast 0. In other words, she ignores all signals, so that actions cannot be chosen based on the social signal that was broadcast. When an agent uses a nonzero signal, its payoff from an interaction is decreased by c .

As formalized here, an agent that does not condition her actions on signals will necessarily also fail to send a signal, and vice versa. In the real world, these may be independent dispositions. A person might abstain from broadcasting a social signal, but still use social signals to determine what action to perform. Conversely, they might broadcast a social signal, but ignore signals from others when choosing what action to perform. Combining these two dispositions so that they always co-occur is computationally convenient, plausible for at least some social contexts, and produces some interesting and novel social outcomes. We leave investigation of a model with independence between signaling and attention for future research.

For this extension, we introduce a third type of agent into the population: the Akkadians (*), who by default make up a third (0.34) of the population. Ummians then ($\square$) make up a proportion $0.33 + \beta$ of the population, with Kishus ($\triangle$) constituting the remainder. We will compare these to simulations without the presence of Akkadians. Recall from our introduction that Akkadians are from a different ethnolinguistic group than the Ummians and Kishus, who are both Sumerian. We therefore model their preferences so that they receive no payoff from coordinating on greetings other than their own. The Kishus and Ummians, however, being threatened by the military might of the Akkadians, receive some payoff from coordinating on their greeting, though not as much as coordinating on another Sumerian greeting (Table 6). Our intention in introducing this third preference type is to consider scenarios in which group identities are nested within one another. For example, Akkadians might perceive the world as divided into two groups—themselves and the Sumerians—whereas the Ummians and Kishus would take umbrage with that delineation, highlighting important differences between their two groups and insisting that the world is instead divided into *three* groups.

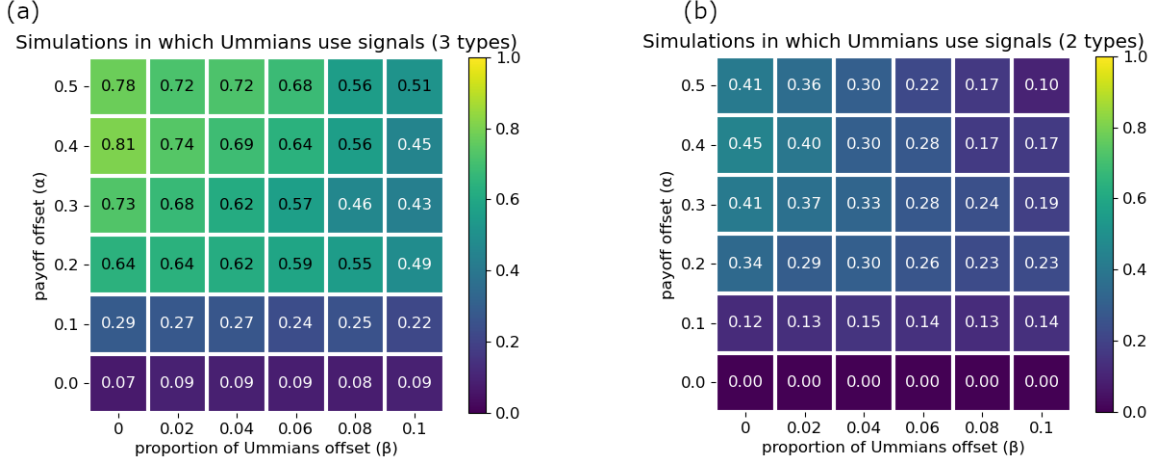
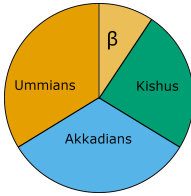


Figure 5: (a) Proportion of outcomes in which Ummians ($\square$) broadcast a social signal, for the model with three types of agents (see Table 6). (b) Proportion of outcomes in which Ummians ($\square$) broadcast a social signal, for the model with two types of agents, in which Ummians constitute $0.5 + \beta$ of the population and Kishus constitute the remainder. In all cases, $c = .01$



Coordination Preferences	Umma greeting	Kish greeting	Akkadian greeting
Ummians ($\square$)	$1 + \alpha$	1	0.5
Kishus ($\triangle$)	1	$1 + \alpha$	0.5
Akkadians ($*$)	0	0	1

Table 6: Coordination payoffs for the model with three preference types.

3.4 Simulation Results: Majority Groups Ignore Social Signals

Our analyses show that the presence of the Akkadians ($*$) can increase the likelihood that the Ummians ($\square$) and Kishus ($\triangle$) use social signals for coordination. Figure 5 shows the proportion of simulations in which Ummians reliably broadcast and attend to signals. When only Ummians and Kishus are present, the Ummians tend to ignore signals and simply use their preferred greeting, particularly when they are in the majority or there is a weaker incentive to coordinate on the Umma rather than the Kish greeting (Figure 5b). In contrast, when Akkadians are present, Ummians are much more strongly incentivized to signal, both because they constitute a smaller proportion of the population, and because the Akkadians are less willing to adopt the Ummian greeting, making it important for the Ummians to condition their actions on their interaction partner. As elsewhere, it is the group that is least willing to budge that drives the rest of the population dynamics (Bergstrom and Lachmann, 2003; O'Connor, 2017).

With three preference types, three action options, and three signaling options, there were many possible outcomes for our simulations. While over half of these rarely occurred, our stochastic simulations yielded a total of thirteen different outcomes that occurred at least 0.5% of the time. All of these outcomes are given a detailed description in Appendix G. Here we focus on the three most prominent outcomes. The first of these is suboptimal, in which

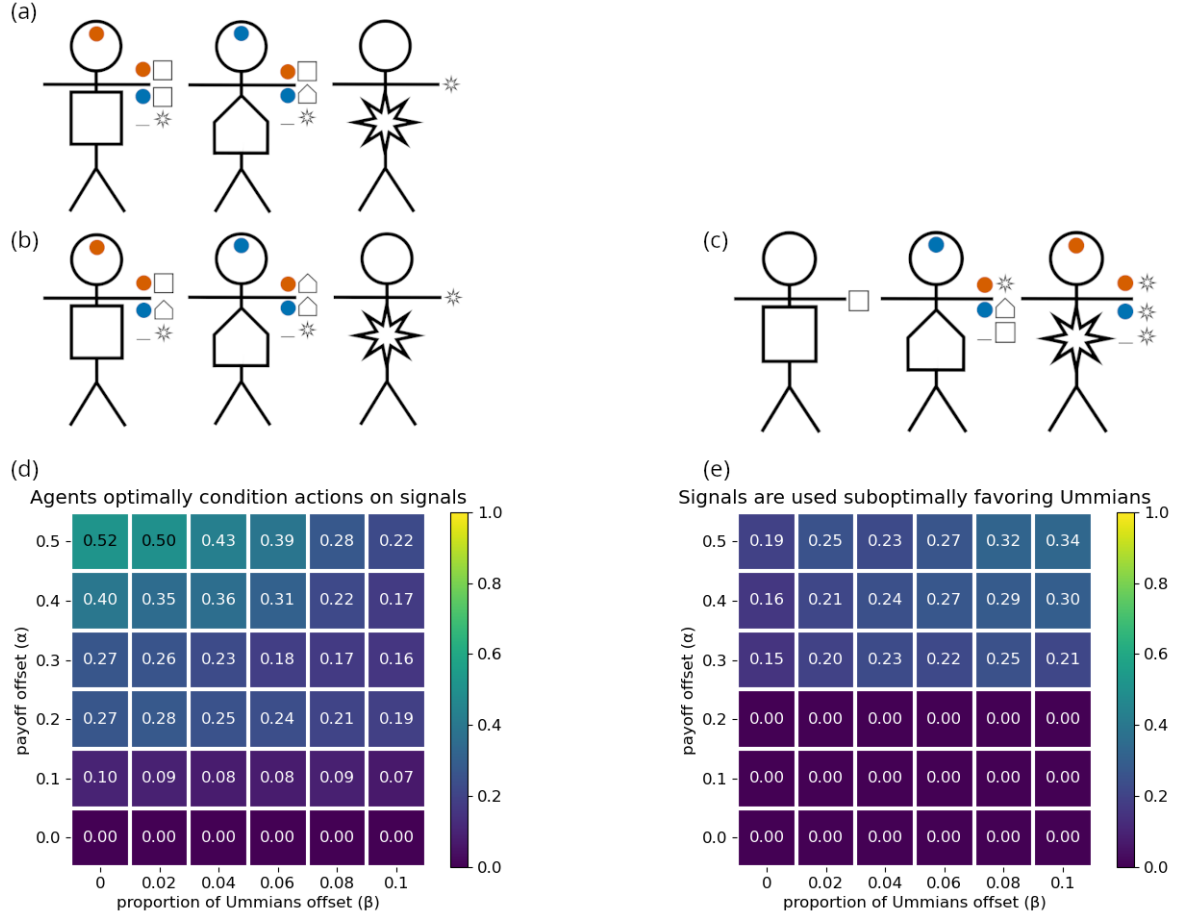


Figure 6: Majority groups ignore social signals. (a) & (b) depict the only two optimal outcomes, while (c) depicts a prominent suboptimal outcome. (d) Proportion of simulations that resulted in either of the two optimal outcomes. (e) Proportion of simulations that resulted in the prominent suboptimal outcome in which Ummians do not use signals.

Ummians (□) and Akkadians (*) always played their respective preference, and Kishus (△) played the Umma greeting with Ummians, the Kish greeting with other Kishus, and the Akkadian greeting with Akkadians. This scenario is illustrated in Figure 6c, in which an agent's signal is depicted in their head, and their actions conditioned on signals received are depicted to the right of their hand. The null signal (0) is represented as an underscore. The prevalence of this outcome is shown in Figure 6e. It is characterized by Ummians (□) signaling 0 and Kishus (△) and Akkadians (*) attending to signals that reliably identified their signaler's type (but with only Kishus conditioning their actions on the signals they received). One can check that this is a Nash equilibrium. It is suboptimal in the sense that when Ummians (□) are paired with Akkadians, there is a failure to coordinate. We observed that there was some path dependency that locked in this outcome. Once sufficient numbers of Ummians evolved to ignore signals, Akkadians faced increased selection to signal in order to allow the Kishus to condition their responses on the Akkadians' signals in order to coordinate more effectively.

The remaining two outcomes highlighted here are optimal in the sense that there were never failures of coordination and agents played their most preferred greeting among them-

selves, illustrated in Figures 6a & b. In both cases, Akkadians ignored signals and always used their preferred greeting, where as both of the Sumerian types successfully used signals to coordinate with Akkadians on the Akkadian greeting, coordinate with their own type on their preferred greeting, and to coordinate with the other Sumerian type with either the Umma or the Kish greeting. The combined frequency of these outcomes is shown in Figure 6d. This scenario was most likely to occur when the Ummians constituted a smaller proportion of the population and when Sumerians had a stronger incentive to coordinate on their own preferred greetings when possible.

Overall, these analysis indicate that larger or more stubborn groups are more often able to ignore social information and simply adopt their preferred actions, while members of minority groups have more incentive to use social signals and to condition their actions on social information. This coheres with the empirical evidence that majority groups, such as whites or heterosexuals in the United States, often do not think of themselves as having any specific social identity and take their norms to be ubiquitous (Simoni and Walters, 2001; Devos and Banaji, 2005). Although this behavioral outcome is optimal in the context of the model, it translates to substantive inequality as minorities face disadvantage in the cognitive load of code switching between norms (Johnson et al., 2022) in addition to the need to often coordinate on their non-preferred norms. If, for the ‘optimal’ outcome favoring Ummians (Figure 6b), we assume each agent type is an equal proportion of the population and $\alpha = 0.5$, then we can calculate the inequality between Ummians, who have an expected payoff of $(1.5 + 1.5 + .05)/3 \approx 1.16$, and Kishus, who have an expected payoff of $(1 + 1.5 + 0.5)/3 = 1$. In other words, an optimal allocation of coordinative behavior does not imply equity.

3.5 Signaling Systems Indicate Affiliations, Actions Capture Power Structures

At this point, we wish to introduce a formal distinction between a *signaling system* and a *group structure*. By “signaling system”, we refer to the system of social signals that agents use to coordinate their actions. A signaling system is fully described when each social signal present in the population has been paired with the subset of all agents in the population that use the social signal. For example, Figure 6a depicts a signaling system in which all Ummians broadcast the red signal and all Kishus broadcast the blue signal. Figure 6b exhibits the exact same signaling system. Signaling systems are often a good first indicator of what groups are present in a population. However, knowing the signaling system used in a population may be insufficient to identify all the groups in the population, as signals may require context or specific knowledge to identify (Smaldino et al., 2018; Smaldino and Turner, 2022). Moreover, knowledge of signaling systems may reveal little about the power dynamics between groups. For example, they do not tell us whether a minority group yields to the norms of the majority. For this, we need information about the actions agents take in response to observing social signals.

We refer to the “structure” of a population to include not only the signaling system but also agents’ full strategy profiles—both the social signals broadcast and the actions they take in response to observing various social signals. For example, while Figures 6a&b show the same signaling system, they show different structures. In Figure 6a, an Ummian gives the Umma greeting when interacting with a Kishu. In Figure 6b, they give the Kish greeting. We can give the same treatment to Figure 4a, which depicts two different structures with the same signaling system. A focus on structures highlights that identity is more than the

signals people use, it is also the actions they take (and expect) in response to those signals. These structures capture what particular signals imply for coordination and power between different groups of people. In some sense, we are arguing against a type of essentialist thinking regarding identity (Smaldino, 2025). While we use preference types as labels to describe the distributions of agents in our model, we also try to take the perspective of the agents themselves and consider how they would categorize the agents in their world. For example, a person who had Kish preferences but the signaling/behavior strategy profile of a typical Ummian would be indistinguishable from other Ummians. This perspective will become especially important in the next section, in which we focus on more complex structures requiring more complex signaling systems.

4 Multidimensional Signaling and Complex Group Structures

In previous sections, our analysis was restricted to one-dimensional, discrete signals of identity. Here we extend the model to allow multiple aspects of identity to be signaled simultaneously to allow for nested or intersectional identities. Individuals can therefore signal multiple identities because they have access to multiple modes of signaling. We will explore the implications of this sort of multidimensional signaling and show how it affords more complex identity structures.

We will continue to use cities and ethnicities from ancient Mesopotamia to designate the different preference types we consider in our model. Here we briefly review our six Mesopotamian groups and summarize their relationships with one another. Each of the four cities depicted in Figure 1 are Sumerian ($\circ$) cities, united by a common language and religious pantheon. Umma ($\square$) and Lagash ($\triangle$) are rivals; they have different local gods and regularly feud over the fertile farmland between the two cities. The temple of the Lagash deity Ningirsu is in the city of Girsu, and we can think of the Girshites ($\diamond$) as a particular devout and zealous offshoot of the Lagashites. Meanwhile, the more distant city of Kish ($\triangle$) traded with both the Ummians and the Lagashites, and so can be considered a neutral third party, sharing features with both of the other Sumerian communities. We will use the Kishus alternately to illustrate (1) a group whose members are simply the conjunction of the people who like the Ummian greeting and those who like the Lagashite greeting, and (2) a group whose members share some preferences with both Ummians and Lagashites but retain distinct preferences of their own. Finally, the Akkadians ($*$) were a non-Sumerian civilization who would eventually go on to conquer all of Mesopotamia. Due to their military superiority, we treat the Akkadians as highly intolerant of the norms of other groups, while Sumerians would tend to be at least somewhat tolerant of the Akkadians.

4.1 The Model with Multiple Signaling Dimensions

We extend the model to allow agents to signal in up to K dimensions. Signaling in each dimension is an independent decision, and for each dimension agents can choose to signal and attend to the signals of others, or else refrain from signaling and attending to signals in that dimension. A person can ignore sports iconography but still be attuned to symbols of religious affiliation. We assume the signal cost c is incurred for each dimension attended to. Adding even one more signaling dimension is sufficient to add considerable complexity. For example, if there are three *de facto* signals (two real signals plus the null signal) in each of

two dimensions, then there are nine different social signals an agent can broadcast. Strategy profiles must therefore include contingent actions for each of the signal combinations an agent can perceive, thus strategy profiles are now strings of length $K + \theta$ where θ is the number of possible signals combinations. To keep things as simple as possible, we will focus primarily on scenarios with $K = 2$, where each dimension has only one signal plus the null signal, for a total of four possible social signals, though we will at times allow for more than one signal possibility in each dimension. Otherwise, the basic replicator dynamics and utility function of the model are the same as described in the previous sections.

Figure 7 illustrates how the stick figure diagrams are extended to accommodate multidimensional signals. Agents' preference types are indicated by the shape on their body, though these are not observable to others. During an interaction, each agent perceives the social signal(s) that their partner broadcasts, depicted on each stick figure's forehead. If an agent does not pay attention to a dimension of social signaling, then no signal for that dimension is shown on the agents' forehead. Finally, an agent's action in response to a given signal is depicted adjacent the given signal near the stick figure's hand. If an agent does not pay the cost of attending to a signaling dimension, then she neither signals in that dimension nor conditions her actions on signals in that dimension. Figure 7c illustrates a scenario where agents fail to coordinate, because each agent is signaling using a dimension that their interaction partner does not attend to. The other subfigures all display scenarios in which the interaction leads to successful coordination.

In the remainder of this section, we will consider how the multidimensional signaling model can represent the identity structures discussed in the Introduction and depicted in Figure 2. Our analysis in this section is more illustrative than thorough, but it will highlight the importance of sufficient modeling complexity in capturing these identity structures and suggest how empirical research using signaling behavior alone may fail to discover deeper identity structures.

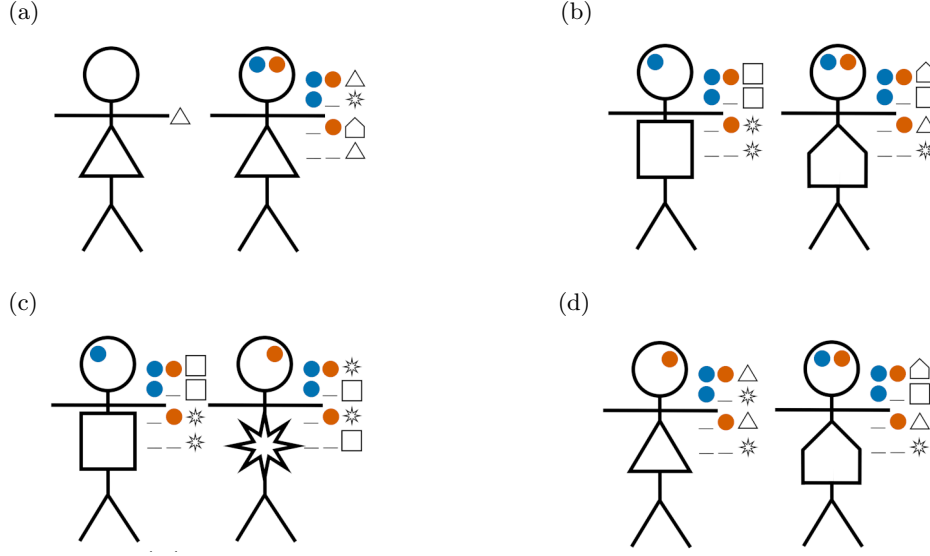
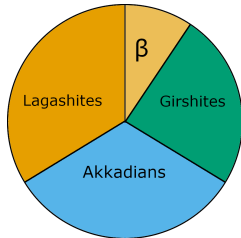


Figure 7: Diagram examples of interactions with multidimensional signals. (a) Successful coordination, both agents give the Lagash greeting (Δ). (b) Successful coordination, both agents give the Umma greeting ($\square$). (c) Agents fail to coordinate, as each attends to a different signaling dimension and the agents have incompatible strategy profiles. (d) Successful coordination, both agents give the Lagash (Δ) greeting.

4.2 The Single Embedding Structure and the Benefit of Multiple Dimensions

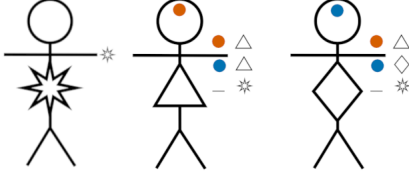
We will begin with a case inspired by the idea of one group being embedded within a larger group, whose members interact with a third distinct group. Of course, we do not model this embedding explicitly; instead, we construct three types of individuals whose preferences are *indicative* of this embedding. Consider the coordination preferences shown in Table 7. Recall that the Girshites can be viewed as a particularly zealous sect of Lagashites. The Girshites have their own distinct greeting on which they most prefer to coordinate. Additionally, given the affinity between Girsu and Lagash, Girshites consider the Lagash greeting the next best thing if they cannot coordinate on the Girsu greeting. Both Girshites and Lagashites will grudgingly adopt the Akkadian greeting if no other option is available, while the Akkadians do not benefit from coordinating on any greeting but their own. We will use this example to show that allowing agents multiple dimensions of social signals makes it much easier for them to find optimal methods of coordination.



Coordination Preferences	Lagash greeting	Girsu greeting	Akkadian greeting
Lagashites (Δ)	1	0	0.5
Girshites ($\diamond$)	1	$1 + \alpha$	0.5
Akkadians ($*$)	0	0	1

Table 7: Coordination preferences for the single embedding subsection.

(a) Optimal Structure
for 1 Dimensional System



(b) Single Embedding Structure

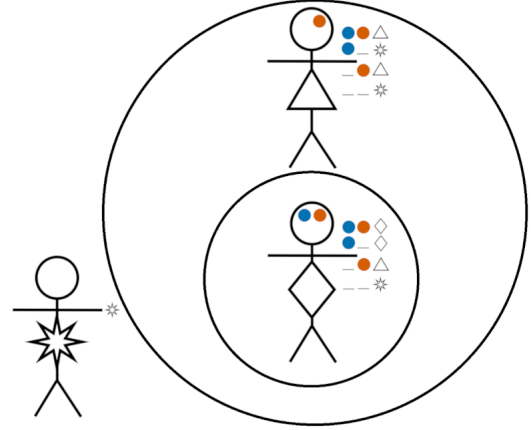


Figure 8: Optimal signaling systems and actions for the coordination preferences shown in Table 7, when agents have access to either (a) one or (b) two signaling dimensions.

We designate a group structure as optimal if the following three conditions are met:

- (i) Given any two agents, if there is some action for which they both receive nonzero payoff, then they always successfully coordinate on an action for which they both have nonzero payoff.
- (ii) Given (i), that action provides the maximal possible sum of the two agents' payoffs.
- (iii) There are no other group structures that meet criteria (i) & (ii) while also allowing agents to pay a lower attention cost.

This definition is tailored to the specific group structures that we investigate. Circumstances not involving coordination among large groups would require different metrics. The scope of our analysis is therefore restricted to dyadic coordination games. It is likely that most or all of the optimal outcomes discussed in Section 4 are Pareto optimal (see Neary, 2012), but our definition is not fully equivalent since there are scenarios in which an outcome that benefits one player without providing a benefit to the other would fail to evolve. For example, suppose there are two preference types such that no action exists for which both types receive nonzero payoff. Then, in such a scenario, our model would produce outcomes in which agents of the two preference types fail to coordinate, given they have no incentive to do so. Such an outcome is not Pareto optimal because you can make a Pareto improvement. This can be done by changing the agents actions such that the two given preference types coordinate on an action for which one type has a nonzero payoff despite the other type having zero payoff for the action. In this case one type would improve their payoff while the other's payoff would not decrease. However, our definition optimality allows us to consider as optimal an outcome in which the two preference types fail to coordinate, given there is no action for which they *both* have nonzero payoffs, and therefore neither type has any incentive to unilaterally change their actions.

In Appendix C we explore an alternative optimality criterion (ii) requiring that an action is considered optimal only if there is no other action for which both agents receive a strictly greater payoff. There are a small number of conditions for this the two definitions give

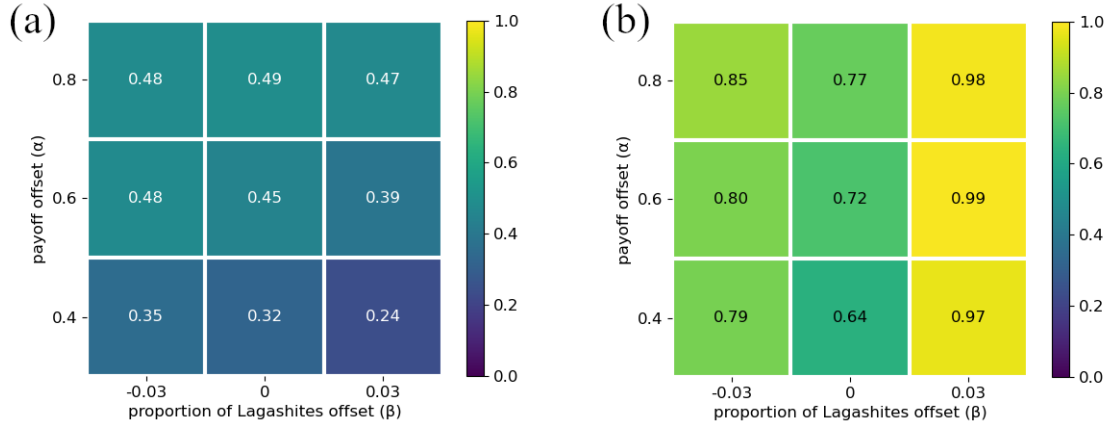
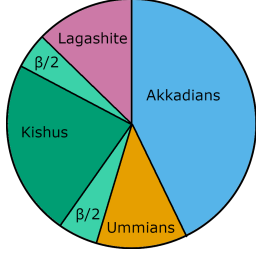


Figure 9: the proportion of simulations that resulted in the optimal outcome when agents had access to either (a) one or (b) two signaling dimensions.

slightly different results. However, for the cases we consider in this paper either criterion is sufficient for identifying, from among the outcomes that are produced in simulations, those outcomes with the highest overall net payoff.

Figure 8 depicts optimal group structures for the coordination preferences given in Table 7 for agents afforded either (a) one signaling dimension with two non-null options or (b) two signaling dimensions with one non-null option in each dimension. By inspection, each of these structures lead to the exact same outcomes for coordination between the possible dyads. Moreover, with two dimensions, the Girshites ending up paying higher attentional costs. Nevertheless, this outcome can be favored over the one-dimensional system. The reason is because it is much more likely that the optimal group structure evolves if agents have access to more than one signaling dimension.

Figure 9 shows the proportion of simulation runs in which agents evolved optimal group structures for (a) one or (b) two signaling dimensions. Our results show that it is substantially easier for agents with multiple signaling dimensions to arrive at an optimal structure. When only allowed one signaling dimension, the Lagashites ($\triangle$) must learn responses to two different social signals: red ($\bullet$) and blue ($\bullet$). When agents are allowed two signaling dimensions, Lagashites only need to learn how to act in response to a single social signal ($\bullet$). Next, consider the Girshites. Suppose the Lagashites have already found a social signal, red ($\bullet$), that allows them to optimally coordinate among themselves. With two signaling dimensions, the Girshites can easily optimize their interactions with the Lagashites by broadcasting the same red signal and giving the Lagash greeting in response to that signal. This does not depend on any change in behavior from the Lagashites. Adding the blue ($\bullet$) signal in a separate dimension allows them to coordinate amongst themselves. As we've already observed, majority preference groups are prone to ignoring the social signals of minority preference groups. With multiple dimensions of social signals, Girshites can broadcast their shared identity with the Lagashites while using a second dimension of signals to optimize coordinate among themselves. This is impossible with one dimensional signals. Multidimensional signals allow the Girshites to signal their shared identity with the Lagashites while the Lagashites ignore the social signal unique to the Girshites, indicative of a niche subgroup embedded within a larger community. More generally, our model suggests that T levels of nested embeddings in the group structure require $T + 1$ dimensions of social



Coordination Preferences	Umma greeting	Kish greeting	Lagash greeting	Akkadian greeting
Ummians ($\square$)	1	0	0	0.5
Kishus ($\triangle$)	1	α	1	0.5
Lagashites ($\triangle$)	0	0	1	0.5
Akkadians ($*$)	0	0	0	1

Table 8: Coordination preferences for the intersectional subsection. Here, Kishus are a proportion $0.2 + \beta$ of the population, Ummians and Lagashites are each $0.2 - \beta/2$ proportion of the population, and Akkadians the remaining 40% of the population.

signals are required.

4.3 The Intersectional Structure and the Importance of Actions

Next, we consider an example inspired by the idea of overlapping or intersecting identity groups, and show how two distinct group structures can exhibit identical signaling systems. Our analysis will highlight how observing only social signals, without accounting for coordinative behaviors, can lead to overlooking intersectional identities.

Consider the coordination preferences shown in Table 8. Recall the narrative that Ummians and Lagashites hate each other, and so receive no payoff from coordinating on each others' greetings. Kishus, on the other hand, are happy to give either the Umma or Lagash greeting. But notice this is compatible with two distinct Kishu identities. Kishus could be a blend of Ummians and Lagashites, happily engaging in either greeting, and without any distinct greeting they prefer to use amongst themselves ($\alpha = 0$). Alternatively, Kishus's preferences may be more than the mere conjunction of Ummian and Lagashite preference. There may instead be a Kish greeting that Kishus prefer above all others ($\alpha = 2$).

The connection to intersectionality should be clear. When the Combahee River Collective formed, it was because there were issues faced by black women that were not being addressed by mainstream feminism nor by Black civil rights movements (Combahee River Collective, 1977). People who were members of the two groups wanted to coordinate in a way that was not reducible to the conjunction of the opportunities for coordination afforded them by either group in isolation. This is clearly a different type of group structure than, for example, a group of people who are both rock climbers and sociologists, who may coordinate on activities related to either group but do not have any additional opportunities for coordination afforded them in virtue of being members of both groups.

As in the previous section, we find that optimal outcomes evolve more frequently when agents are allowed multidimensional signals (see Appendix I). Here we only show results for simulations in which agents were allowed multidimensional signals, one signal in each of two dimensions. For all parameters shown here, at least 95% of the simulations resulted in the conjunctive signaling system (Figure 10). When Kishus did not have a greeting that they preferred over the Umma and Lagash greetings ($\alpha = 0$), they evolved to use one of these amongst themselves. The proportion of simulations that resulted in either of these two group structures is shown in Figure 11b. When Kishus did have a strong preference for a distinct greeting ($\alpha = 2$), then the optimal structure is the intersectional structure shown in Figure 10b. The proportion of simulations that resulted in this structure is shown in 11a.

Our interpretation of intersectional identity aligns with the suggestion by Varley and

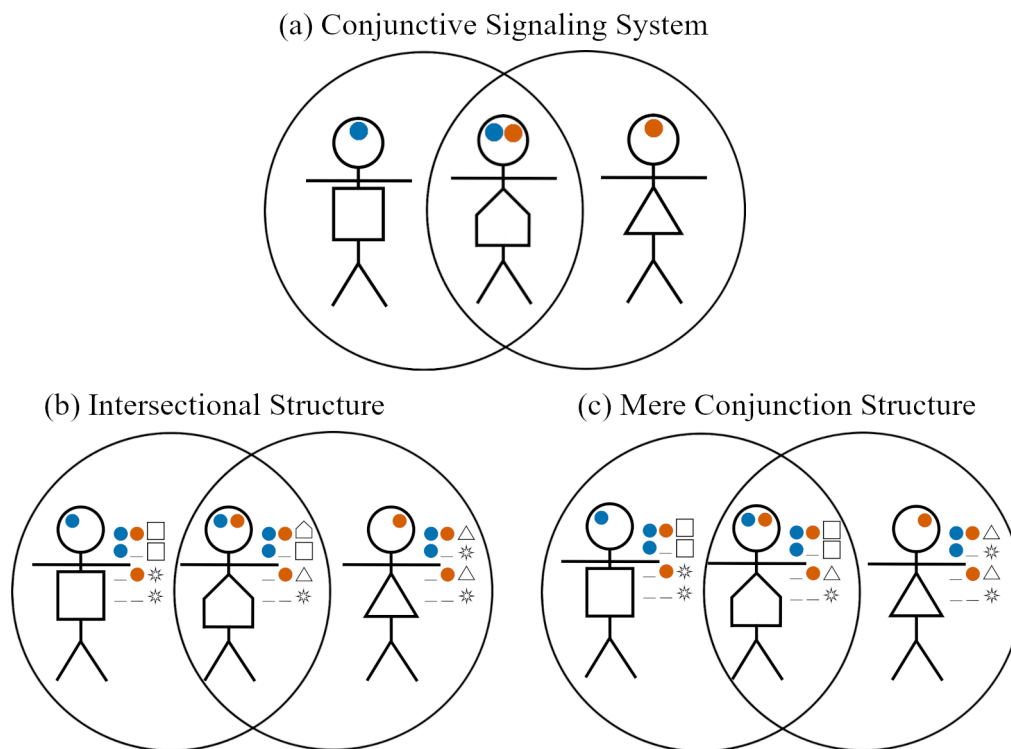


Figure 10: The conjunctive signaling system (a) can be indicative of two distinct group structures: one in which Kishus use their own preferred greeting among themselves (b), and one in which they use the Ummian greeting (c).

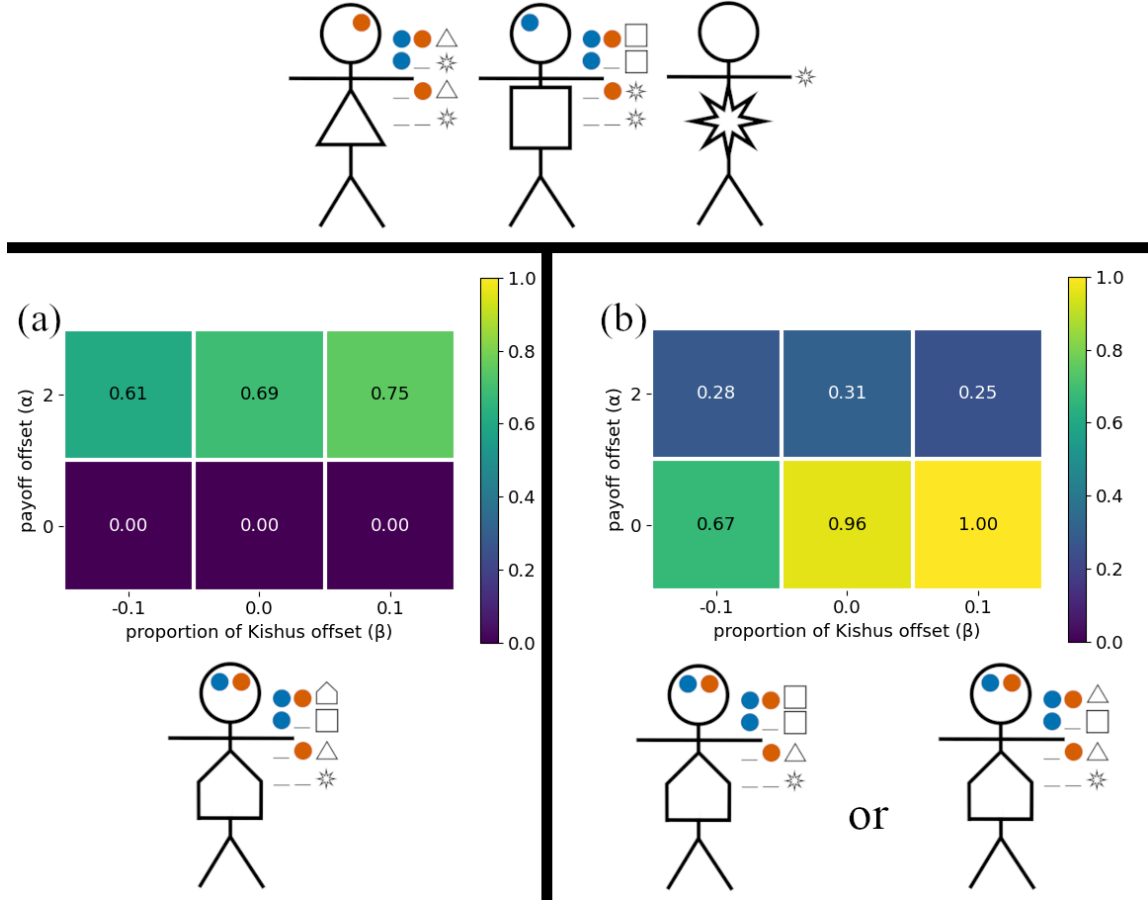


Figure 11: Proportion of simulations that resulted in (a) the intersectional structure, optimal when $\alpha = 2$, or (b) either of the two mere conjunction structures, optimal when $\alpha = 0$. The trio of stick figures at the top represent the optimal strategy profiles used by non-Kishus in all cases.

Kaminski (2022) that intersectional identities can be identified as having a greater proportion of total information being synergistic information. Synergistic information is a formal measure of the information gained when considering the combined effect of multiple sources of information, in this case information from what was signaled in the first dimension combined with information from what was signaled in the second dimension. Using partial information decomposition (Williams and Beer, 2010) and focusing on Varley and Kaminski’s definition of synergistic information, we found that, for $\beta = 0$, the synergistic information in the intersectional equilibrium, (Figure 11a) was around 49% of the total information. Compared to this, the synergistic information in the equilibria for which Kishus were merely members of both groups (Figure 11b) averaged only 36% of the total information.

Figure 11 shows that simulations frequently resulted in optimal outcomes. However, optimal outcomes became less frequent when Kishus constituted a smaller proportion of the population. We observed in Section 3.4 that large majority preference groups can evolve to ignore social signals, while minority preference groups evolve to attend to them. Here we see a limit to this effect. When Kishus are a smaller proportion of the population,

they have fewer opportunities to learn the relevant response to conjunctive social signals to obtain the optimal structure. This suggests that in the real world, there may be minority intersectional identities (with respect to individuals' preferences) that we are simply unable to identify because they are too small for their members to learn the strategy profiles that would identify them as an intersectional group. We note that this result is obtained in the absence of ingroup homophily, which could provide more opportunities to learn group-relevant signals, but which would also require these potential intersectional groups to have some mechanism for assortment other than signaling.

If we assume that people exhibit optimal behavior given their preferences, it is likely that their social signals will underdetermine the population's structure. Therefore, if empirical researchers only consider the group labels or social signals that people use, then an intersectional structure and a mere conjunction structure will look identical. Furthermore, a biased sampling of interaction pairings can skew inference. In the conjunctive signaling system, for example, Kishus look like Lagashites from the perspective of Lagashites and like Ummians from the perspective of Ummians. This gives more reason to think we are prone to overlook some identities.

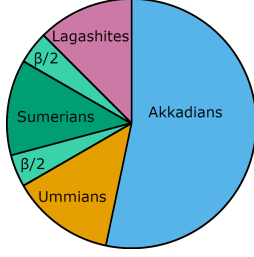
To overcome these issues, attention must be paid to how people coordinate their actions with different partners. This involves anticipating unique group-specific behavioral domains by conducting inductive assays on what individuals hope others say, react, or do during an interaction. Our model shows this is especially important in an intersectional group, such that a bottom-up approach to documenting coordination foci is imperative across the range of group membership.

4.4 Double Embedding Structures

We refer to scenarios in which more than one subgroup is embedded in a larger group as double embedding structures. Here we compare two different types of double embedding group structures, and illustrate the fact that there are social structures in nature that cannot be adequately modeled without agents being afforded sufficiently complex social signals.

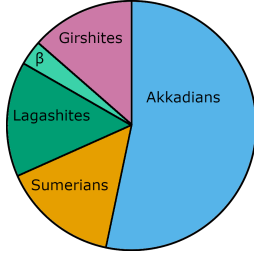
Consider the coordination preferences in Table 9. While Ummians and Lagashites get no benefit from each other's most preferred greeting, they are both Sumerian and so prefer a generic Sumerian greeting over the Akkadian greeting. In this scenario, we allow agents signals in two dimensions such that they have access to one non-null signal in the first dimension and two non-null signals in the second. In the optimal structure, this will allow agents to use that one signal in the first dimension to indicate the Sumerian identity that is shared between Ummians, Lagashites, and Sumerians at large; and, it will allow Ummians and Lagashites to signal their differences by using different non-null signals in the second dimension. We call this the disjoint double embedding structure (Figure 12, right) and it is the unique optimal outcome. Figure 12 (left) shows that the disjoint double embedding obtained in about half the evolutionary simulations we ran.

Now consider the coordination preferences in Table 10. Recall that the Girshites can be viewed as a sect of particularly zealous Lagashites. The Girshites have their own distinct greeting which they most prefer to coordinate on. Girshites also prefer the Lagash greeting over the generic Sumerian greeting, which in turn is preferred over the greeting of the foreign Akkadians. In this scenario, we allow agents one non-null signal in each of three dimensions. Allowing agents three dimensions of signals allows three degrees of identity specificity to be broadcast at once. It allows Girshites to broadcast a signal specific to themselves in one dimension, a signal shared with the Lagashites in another, and a signal



Coordination Preferences	Sumerian greeting	Umma greeting	Lagash greeting	Akkadian greeting
Sumerians ($\odot$)	1	0	0	0.25
Ummians ($\square$)	1	$1 + \alpha$	0	0.25
Lagashites ($\triangle$)	1	0	$1 + \alpha$	0.25
Akkadians ($*$)	0	0	0	1

Table 9: Coordination preferences for the subsection on the disjoint double embedding. In simulations, Sumerians were $0.15 + \beta$ proportion of the population, Ummians $0.15 - 0.5 \times \beta$ proportion of the population, Lagashites $0.15 - 0.5 \times \beta$ proportion of the population, and Akkadians the remaining 55% of the population. Agents were allowed one signal in the first dimension and two in the second.



Coordination Preferences	Sumerian greeting	Lagash greeting	Girsu greeting	Akkadian greeting
Sumerians ($\odot$)	1	0	0	0.25
Lagashites ($\triangle$)	1	$1 + 0.5 \times \alpha$	0	0.25
Girshites ($\diamond$)	1	$1 + 0.5 \times \alpha$	$1 + \alpha$	0.25
Akkadians ($*$)	0	0	0	1

Table 10: Coordination preferences for the hierarchical double embedding. In simulations, Sumerians were 0.15 proportion of the population, Lagashites $0.15 + \beta$ proportion of the population, Girshites $0.15 - \beta$ proportion of the population, and Akkadians the remaining 55% of the population. Agents were allowed one signal in each of three dimensions.

shared with all Sumerians in a third dimension. So Sumerians can optimally coordinate by recognizing the Sumerian social signal broadcast by Lagashites and Girshites without having to know anything about the signals that are specific to Lagashites and Girshites. Likewise, Lagashites can optimally coordinate with Girshites by recognizing the Lagashite signal without having to know about anything specific to the Girshite identity. And Girshites can optimally coordinate among themselves using their own signal. Given agents preferences, the hierarchical double embedding structure (Figure 13, right), is the unique optimal outcome. Figure 13 (left) shows that the hierarchical double embedding obtained in about half the evolutionary simulations we ran.

Both double embedding group structures showcase complex signaling and behavioral phenomena that could not be captured by a simpler model with only one dimension of social signals. It is not just the case that multidimensional signals make it easier for agents to arrive at optimal outcomes. In addition, the *ways* that they arrive at optimal outcomes depends on their signaling affordances. In the disjoint double embedding structure, Ummians and Lagashites just look like Sumerians from the perspective of Sumerians, because they all broadcast the same social signal. This cannot be represented in a model with just one dimension of social signals. In the hierarchical double embedding, from the perspective of the Sumerians, Lagashites and Girshites look like Sumerians because of their shared social signal; Girshites look like Lagashites to the Lagashites, while from the perspective of Girshites, they are distinguishable from the two groups. Exhibiting these three perspectives in a single structure is only possible with at least three dimensions of social signals.

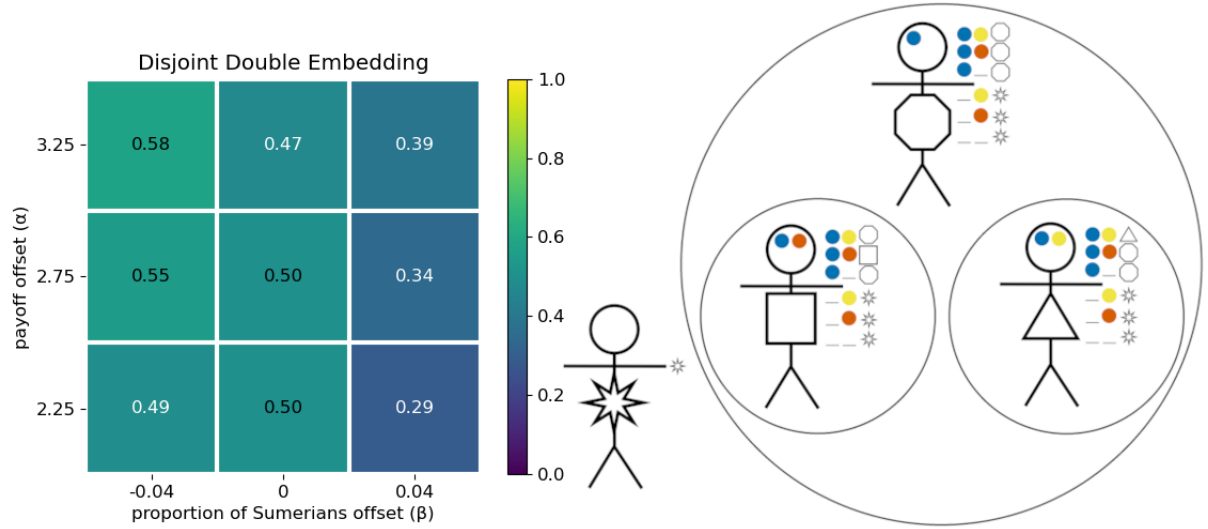


Figure 12: Proportion of simulations that resulted in the disjoint double embedding structure.

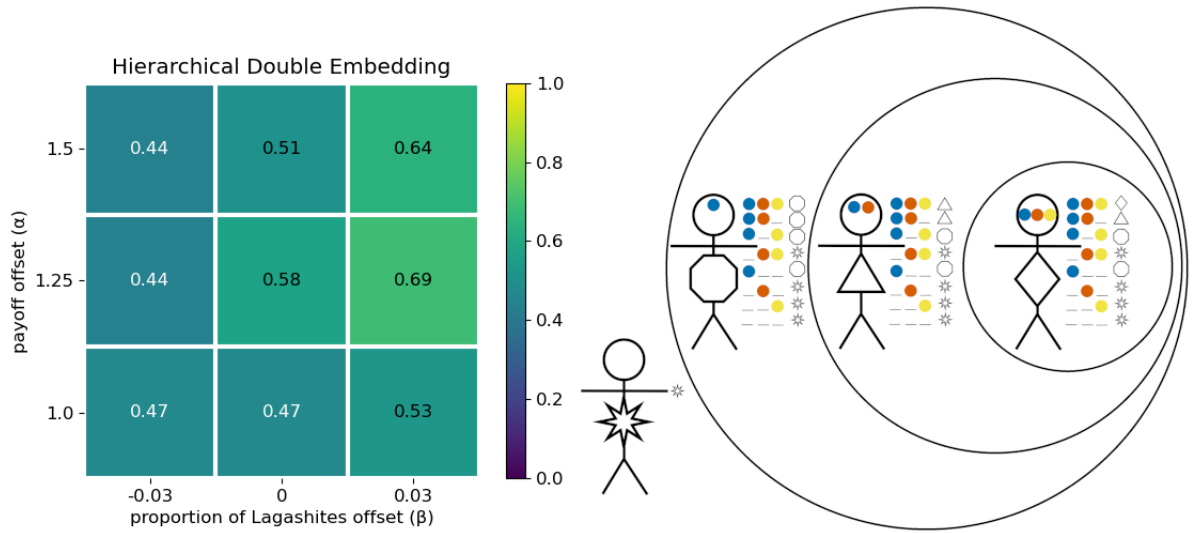


Figure 13: Proportion of simulations that resulted in the hierarchical double embedding structure.

5 Discussion

We have considered the nature of identity signaling in societies with diverse preferences for coordination. Our analysis provides a foundation for a general theory of identity signaling in the context of coordinating normative behavior in a structured population, including the role of multidimensional signaling in affording complex group structures involving nested and overlapping identities with unequal power.

Without social signals, interactions in diverse societies will tend to be suboptimal, with this suboptimality manifesting in at least two ways. First, a society could be characterized by failures of coordination whenever people holding different preferences interact. In this case, because people lack social signals to identify the type of individual they are interacting with, they may simply adopt the action they most prefer. This scenario is most probable when preferences are strongly held or when a society is very diverse and no preference is held by a significant majority. Second, interactions will be suboptimal when either preferences are weakly held or when a significant majority of the population holds the same preferences. In this latter case, a dominant norm will tend to emerge on which people always successfully coordinate. However, this successful coordination comes at the cost of some people, typically members of minority groups, adopting an action for coordination that they do not like, even when interacting among themselves. Our analysis replicates prior research suggesting that minorities have low odds of maintaining their culture when frequently interacting with members of a dominate culture (O'Connor, 2019; Bunce and McElreath, 2018). Many prior evolutionary models of ethnic markers focus on their use in partner selection, assuming that individuals always use the same norm once an interaction has begun (Boyd and Richerson, 1987; McElreath et al., 2003; O'Connor, 2019; Bunce and McElreath, 2018). If only individuals could choose their actions based on whether they were interacting with an in-group or out-group member, minorities could coordinate on their more preferred action among themselves despite being forced to adopt the majorities preference for between group interactions.

By adopting social signals, individuals can avoid these suboptimal situations by conditioning their actions on the signals of the person they are interacting with (Section 3.2). Here, our model closely aligns with prior work by Bunce (2021), who found that cross-cultural competence allows minorities to sustain their culture. Strategic use of social signals allow members of a minority group to adopt the majority's preference as the norm for coordination between groups while retaining their preferred norms when interacting among themselves. Bunce (2020) finds empirical support for this idea in his studies of intra- and inter-ethnic interactions in Peru, showing that cross-cultural competence is most prevalent among minorities living among a larger group.

Previous research has indicated that social signals—also called identity signals, tags, or ethnic markers—are particularly useful at borders where different cultures meet (Barth, 1969; Boyd and Richerson, 1987; McElreath et al., 2003; Johnson et al., 2024). While our model does not include a spatial or network component, the random mixing between agent types is reflective of what one might expect at a border between cultures rather than at the center of a homogeneous group. It may also represent more quotidian interactions in a diverse population under moderate or low amounts of segregation. When there is uniformity across a population there is no need for social signals. In fact, majority groups often act *as if* there is uniformity rather than diversity in the population by unflinchingly taking their preference to be the norm (Simoni and Walters, 2001; Devos and Banaji, 2005). Our model produces this result, and compliments previous models of ethnic markers by focusing on

using signals for selecting behaviors during an interaction that has already begun.

A reality of modern society is that people have complex social identities (Roccas and Brewer, 2002; Smaldino, 2019). Who we are is often determined, at least in part, by *where* we are and whom we are with. There is therefore a significant benefit to the ability to independently signal different aspects of one’s identity, so that one aspect can be detected even if other aspects go ignored or unappreciated, as when an individual belongs to a niche group that is embedded within a larger group. Members of the larger group can optimize their behavior without having to learn the social signals associated with the embedded group, while members of the smaller group can still coordinate optimally when the opportunity arises. We have demonstrated how the ability to signal in multiple dimensions simultaneously creates the conditions for hierarchical or intersectional identity signaling to emerge and achieve optimal coordination. Relatedly, we also demonstrated how considering the ways in which different individuals coordinate (or fail to coordinate) is just as important as considering the signals they send to determine the structure of identity and power relationships in a population. For example we showed in Section 4.3 that the same patterns of signaling could be associated with individuals that are equally affiliated with each of two groups, or with individuals whose intersectional identities are not captured by the conjunction of their other affiliations. Only by examining their behaviors can the difference between these two signaling systems be parsed. This has implications for empirical researchers interested in identity, particularly those interested in capturing community relationships based solely on the signals used by the constituents of those communities. A focus on signals alone will fail to capture the group structures that include nested or intersectional identities as well as power dynamics.

We believe our model is a good fit for some aspects of coordination among people with intersectional identities. However, it should be noted that intersectionality is typically discussed in terms of disadvantage experienced by marginalized groups. What we have called the intersectional structure requires only that there is some group whose members signal two identities but coordinate among themselves on an action that is not preferred by people who are members of only one of the two groups. This could occur among a group of people that does not experience any of the disadvantages traditionally associated with intersectionality. Readers for whom intersectionality does not exist without this power imbalance may wish to restrict use of the term “intersectional” in the context of the model only for those group structures that include similar power imbalances, with “non-reductive group structures” being a more general term that does not imply a particular power relationships.

Our model captures the three step process of (i) differences in *preferences* (ii) contributing to the evolution of social *signals* such that (iii) people can optimally condition their *actions* based on identifying in-group and out-group members through their social signals. In so doing, we encapsulate more of the complexity of human identity signaling and coordinative behavior than most related modeling work. Nevertheless, there are important limitations that can and should be addressed to produce a more comprehensive theory of identity signaling in diverse populations. First, we do not address how people come to have different preferences in the first place. Preferences are often shaped by our social groups and identity commitments (Cialdini and Goldstein, 2004; Fehr and Hoff, 2011), so an important direction for future research is to consider how a model of optimal signaling and assortment coheres with models of identity-influenced preference change (Steiglechner et al., 2023, 2025), which might additionally include the role of immigration and even long-distance interaction (Kunst and Mesoudi, 2025; Smaldino et al., 2025). In a future model, we hope to include dynamic preferences that evolve concurrently with group signals and

individuals' dispositions to act in particular ways in response to social signals. Second, we limited the context for interactions (and therefore preferences) to coordination, as captured by the generalized Bach or Stravinsky game. While coordination is at the heart of much of human social behavior (Bicchieri, 2005; Calcott, 2008; Eickers, 2023), other contexts are better captured by other games, including games involving more than two players. For example, Bednar and Page (2007) have advocated for the importance of exploring contexts in which defecting from a cooperative strategy is incentivized and we may want to explore contexts in which such defection are conditioned on complex group memberships. Third, we only considered discrete, all-or-none signals. While some signals, such as symbols or words may be discrete, others contain more continuity. For example, consider the strength or consistency of an accent in spoken language (Cohen, 2012). In the United States, a slight drawl might signal someone is from the South, while a stronger accent might addi- tional signal that they are from a particularly rural part of the South. Finally, we only considered overt signals that are unambiguous and equally observable to any interlocutor. However, many signals of identity are encrypted or covert, especially those used by minor- ity groups, oppressed groups, or groups incentivized to maintain secrecy (Smaldino et al., 2018; Smaldino and Turner, 2022). Such covert signals have not yet been considered in the context of complex identity structures with multidimensional signaling. While we have yet to explore these future research directions, we have laid a solid foundation for these and other possible continuations of modeling social identities.

Figure Captions

1. (a) Ancient Sumerian cities used throughout the paper as a motivating example. (b) Diagram of relationships between the different Mesopotamian identities along with shapes used to represent each of them throughout the paper.
2. A rough sketch of some group structures that can be produced with our model.
3. (a) Proportion of simulations in which agents always give their preferred greeting. (b) Proportion of simulations in which everyone gives Ummians' preferred greeting.
4. (a) Proportion of outcomes that are one of the two optimal outcomes. (b) Proportion of outcomes that are the optimal outcome favoring the Ummians.
5. (a) Proportion of outcomes in which Ummians ($\square$) broadcast a social signal, for the model with three types of agents (see Table 6). (b) Proportion of outcomes in which Ummians ($\square$) broadcast a social signal, for the model with two types of agents, in which Ummians constitute $0.5 + \beta$ of the population and Kishus constitute the remainder. In all cases, $c = .01$
6. Majority groups ignore social signals. (a) & (b) depict the only two optimal outcomes, while (c) depicts a prominent suboptimal outcome. (d) Proportion of simulations that resulted in either of the two optimal outcomes. (e) Proportion of simulations that resulted in the prominent suboptimal outcome in which Ummians do not use signals.
7. Diagram examples of interactions with multidimensional signals. (a) Successful coordination, both agents give the Lagash greeting ($\triangle$). (b) Successful coordination, both agents give the Umma greeting ($\square$). (c) Agents fail to coordinate, as each attends to a

- 879 different signaling dimension and the agents have incompatible strategy profiles. (d)
Successful coordination, both agents give the Lagash (Δ) greeting.
- 882 8. Optimal signaling systems and actions for the coordination preferences shown in Table 7, when agents have access to either (a) one or (b) two signaling dimensions.
9. The proportion of simulations that resulted in the optimal outcome when agents had access to either (a) one or (b) two signaling dimensions.
- 885 10. The conjunctive signaling system can be indicative of two distinct group structures: one in which Kishus use their own preferred greeting among themselves (b), and one in which they use the Ummian greeting (c).
- 888 11. (a) Proportion of simulations that resulted in (a) the intersectional structure, optimal when $\alpha = 2$, or (b) either of the two mere conjunction structures, optimal when $\alpha = 0$. The trio of stick figures at the top represent the optimal strategy profiles used
891 by non-Kishus in all cases.
12. Proportion of simulations that resulted in the disjoint double embedding structure.
- 894 13. Proportion of simulations that resulted in the hierarchical double embedding structure.

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Author Contributions

900 PES and AVB formulated the original research questions. NG and PES refined these questions and designed the model. NG wrote the model code and analyzed the outputs of the simulations. NG and PES wrote the paper initial draft of the paper. NG, PES, and AVB
903 edited the paper revisions.

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Conflicts of Interest

909 n/a

Research Transparency and Reproducibility

https://github.com/nathanlgabriel/social_identity_signaling

Data Availability

https://github.com/nathanlgabriel/social_identity_signaling

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